

42.510 Statistical Learning for Data Science

Model assessment and selection I

Bikramjit Das

Week 5, Lecture 1

Announcement

- Midterm exam: 21st October, 7pm - 9pm.
- Venue: Think Tank 14/15 (1.509).
- Syllabus: Material from Weeks 1 - 5.
- Paper-based exam. Around 40% - 50% programming/ R-based.
- Please upload .R or .Rmd file via e-dimension (link will be available in folder: Mid-term).
- Bring your laptop with R/R-studio installed along with packages installed during class/exercise.
- Data will be made available via e-dimension (folder: Mid-term).
- Open class notes, open book (referenced text).
- You are not allowed to use generative AI tools or search in the internet
 - ▶ Any violation will result in a zero score.
- Please clarify doubts via Piazza (until Monday night).
- I will keep additional office hours 5 pm - 6 pm this Friday on Zoom.

Outline

- Model assessment and model selection
- Bias – Variance trade-off
- Subset selection
- Cross validation
- LASSO

Model assessment and model selection

- We will use ideas in regression and classification developed in the last couple of decades.
- These were mostly developed for large data sets with many predictors (also primarily developed before the current information revolution).
- Primary goal
 - ▶ prediction accuracy
 - ▶ model interpretability

Large data sets



- n : # observations.
- p : predictors, attributes.
- Classical statistics: $n \gg p$.
- But sometimes: $n \sim p$ or $n \ll p$.
- Cancer dataset
 - ▶ many many genes (potential predictors).
 - ▶ a very small sample.
- High flexibility in model selection.

Bias-Variance trade-off

- Think of a linear regression model. The true model is:

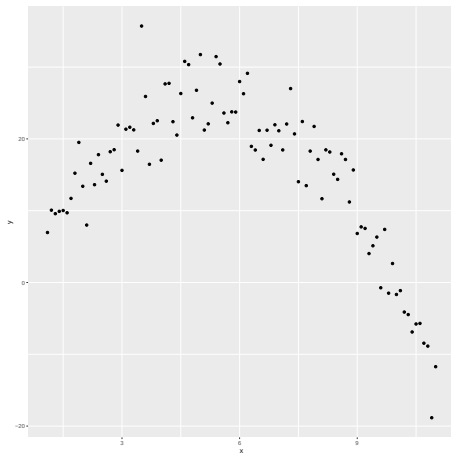
$$Y = f(X) + \epsilon$$

where ϵ is a random error term with mean 0 and variance σ^2 .

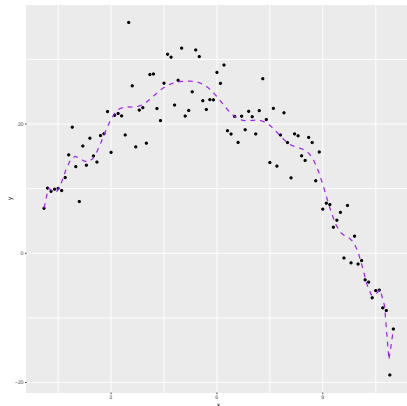
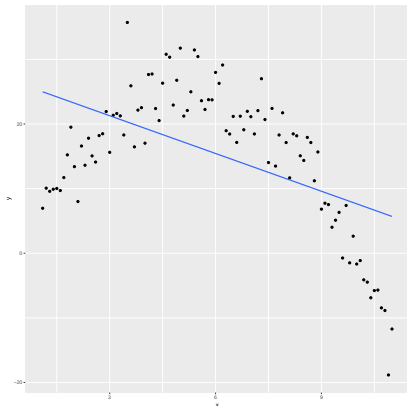
- Suppose we use least squares minimization to find predictor $\hat{f}$ for f .
- We use training set data to find $\hat{f}$ but expect good performance out-of-sample.
- Let (X_0, Y_0) be a (test) data point.
- We want low test MSE or low test RSS (residual sum of squares):

$$\text{Test MSE} = \mathbb{E}(Y_0 - \hat{f}(X_0))^2.$$

Balancing bias and variance

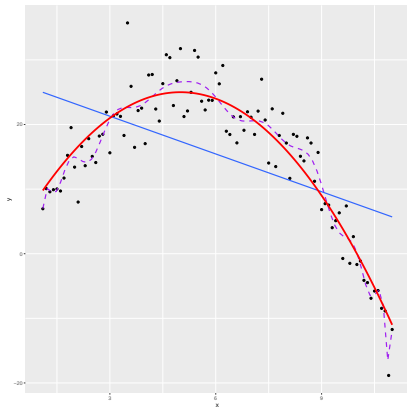


Balancing bias and variance



- Left: Linear regression fit.
- Right: Higher degree polynomial fit.

Balancing bias and variance



- Blue line: linear regression.
- Purple line: higher degree polynomial.
- Red curve: actual function.

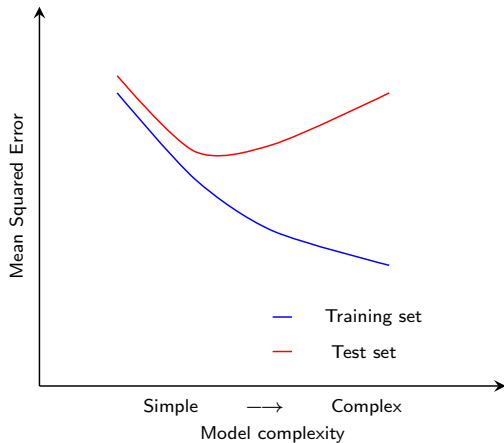
Bias-Variance trade-off

- Test set MSE can be written as:

$$\begin{aligned}\text{Test set MSE} &= \mathbb{E}(Y_0 - \hat{f}(X_0))^2 \\ &= \text{Var}(\hat{f}(X_0)) + [\mathbb{E}(\hat{f}(X_0)) - f(X_0)]^2 + \sigma^2 \\ &= \text{Variance of estimator} \\ &\quad + \text{Squared Bias} \\ &\quad + \text{Variance of error term (irreducible error)}.\end{aligned}$$

- Complex model: typically high variance and low bias.
- Simple model: Low variance but high bias.
- Find the right balance.

Training set error and test set error



Subset selection

- We have n observations, and p predictor variables.
- If $n \approx p$ or $n < p$: risk of over-fitting.
- Pick a selection of important explanatory variable from the p available.
- Solution space: 2^p subsets.

Best subset selection

- Let M_0 denote a null model with no predictors (only the intercept).
- For $k = 1, \dots, p$, do:
 - ▶ Fit all $\binom{p}{k}$ models that contains exactly k predictors and the intercept.
 - ▶ Pick the best among these models by choosing the model with the minimum sum of squared errors for linear regression or maximum log-likelihood for logistic regression. Call this model M_k .
- Choose the best model among $M_0, M_1, \dots, M_p$ using cross-validation errors or adjusted R-squared (linear regression) or AIC (logistic regression).
- Complexity is $O(2^p)$: we need to solve 2^p linear or logistic regressions.

Forward stepwise selection

- Let M_0 denote a null model with no predictors (only the intercept).
- For $k = 1, \dots, p$, do:
 - ▶ Fit all models that augment the predictors in model M_{k-1} with exactly one more predictor (a total of $p - k + 1$ models are fit in step k).
 - ▶ Pick the best among these models by choosing the model with the minimum sum of squared errors for linear regression or maximum log-likelihood for logistic regression. Call this model M_k .
- Choose the best model among $M_0, M_1, \dots, M_p$ using cross-validation errors or adjusted R-squared (linear regression) or AIC (logistic regression).
- Complexity is $O(p^2)$: but there is no guarantee that this will always find the optimal solution.

Backward stepwise selection

- We start with the subset including all variables.
- We drop variables one at a time rather than adding.
- Has the same complexity as forward stepwise selection.
- Solutions from backward & forward methods can be different.
- Solutions of both can differ from the best subset selection solution.
- Complexity is $O(p^2)$: no guarantee of optimality.

Cross-Validation

- Model assessment technique.
- A model is considered good if it has a low *test set error* (*TEST MSE*) .
- We often do not have a large test set to validate our model.
- One method of model assessment here is *Cross Validation*.

Validation set approach

- Divide the data randomly into 2 subsets (often roughly of equal size): the *training set* and the *validation set* or *hold-out set*.
- Use the training set to fit the model, and the validation set to predict and then estimate Mean squared error (MSE).

Potential drawbacks

- The method depends on the points chosen, hence different choices may lead to starkly different estimated MSEs.
- Since we are only using a subset of the available data set, the performance of the model is worse than it would be on a larger data set. And the error estimates tend to be larger.

LOOCV: Leave one out cross validation

Compensates for the drawbacks of the *Validation set approach* yet keeping the same spirit.

- For every $i \in I = \{1, \dots, n\}$, train the model on the set $I \setminus \{i\}$.
- Use this model to predict the i th response, say it is $\hat{y}_i$; and then compute $\text{MSE}_i = (y_i - \hat{y}_i)^2$.
- Compute cross validation error

$$CV_{(n)} = \frac{1}{n} \sum_{i=1}^n \text{MSE}_i.$$

Advantages

- Less biased, since we are fitting the model to $n - 1$ of the points.
- Doesn't change depending on the random sample, like validation set method.

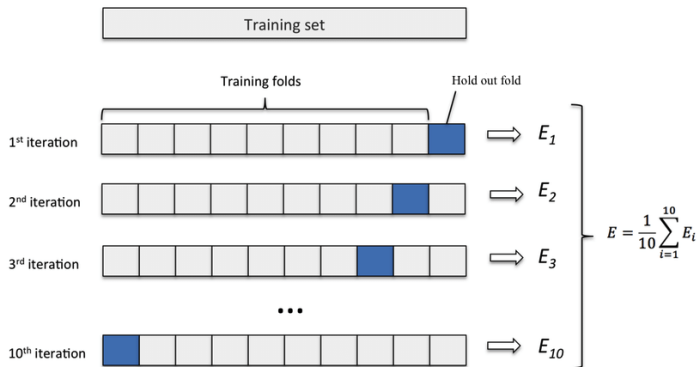
Potential drawback

Computationally intensive: we need to fit n models.

k -fold cross validation

- Divide the data randomly into k subsets (folds) of (roughly) equal size.
- Start with the first fold as a validation set and use the remaining $k - 1$ folds to fit the model.
- Compute the error of the fitted model in the held-out fold.
- Repeat steps 2 and 3 by using the second, third and so on folds as the hold-out fold with the remaining $k - 1$ folds to fit the model.
- Average the error across all the k fitted models to estimate the cross-validation error.
- Some of the commonly used choices for k are 5 or 10.
- When $k = n$, this reduces to LOOCV.

k-fold cross validation



References

- Hastie et al. (2001), *The Elements of Statistical Learning*, Chapters 3,7.
- James et al. (2023), *Introduction to Statistical Learning*, Chapters 5,6.